

SMM069 Advanced Predictive Analytics

Bayes Business School, City St. George's, University of London

Camilla Ayres [250061774]

Ramprashanth Elangovan [250055512]

Mahad Khan [250020895]

Raghav Iyengar [250057230]

Coursework 1: Shrinkage Estimation for Portfolio Construction

A1: Data Preprocessing

We use `DA441_weekly_returns.csv` (441 S&P 500 stocks, 1,251 weekly observations spanning 2000–2023) and `risk_free_rate_weekly.csv` and we find:

(i) **Risk-free rate: percent $\rightarrow$ decimal.** The risk-free series is provided in percentage form and is divided by 100,

$$r_{f,t}^{\text{dec}} = \frac{r_{f,t}^{\%}}{100}.$$

(ii) **Weekly excess returns.** After fixing our 100 stocks with `set.seed(7)` and drawing them from the 441 available, the weekly excess return of stock i in week t is

$$r_{i,t}^{\text{excess}} = r_{i,t} - r_{f,t}^{\text{dec}},$$

giving a $1,251 \times 100$ matrix of excess returns.

A2: Selection of Estimation Models

Mean vector estimators (all used). Sample mean, St-MSh, D-MSh, O-LSh, T-LSh, Wang, BOP and Jorion. We use the Jorion estimator rather than CW because our training window is $n = 156$ weeks against $p = 100$ stocks, which is the $n > p$ regime for which Jorion is defined. CW is the $p > n$ extension of James–Stein and uses the Moore–Penrose pseudo-inverse $\hat{\Sigma}^+$. When $n > p$ the sample covariance is non-singular, so $\hat{\Sigma}\hat{\Sigma}^+ = I_p$, the term $(I_p - \hat{\Sigma}\hat{\Sigma}^+)\bar{X}$ vanishes, and CW collapses to a scalar multiple of the sample mean. It therefore returns the same Sharpe ratios as the Sample estimator, so we exclude it.

Covariance estimator (one chosen): cov1Para. We select the single-parameter Ledoit–Wolf shrinkage estimator that shrinks the sample covariance toward a scaled identity target,

$$\hat{\Sigma} = \delta \mu I_p + (1 - \delta) S, \quad \mu = \frac{1}{p} \text{tr}(S),$$

with $\delta \in [0, 1]$ estimated from the data. With only 156 weekly observations to estimate a 100×100 covariance matrix, the sample covariance S is poorly conditioned and its inverse, which enters every portfolio weight, is highly unstable. `cov1Para` guarantees a well-conditioned, invertible $\hat{\Sigma}$, reducing estimation error and improving out-of-sample stability.

A3: Portfolio Construction and Rolling Window Setup

Rolling windows. Training window = 156 weeks (3 years), testing window = 4 weeks (1 month), step size = 4 weeks. The number of windows is

$$W = \left\lfloor \frac{N - 156 - 4}{4} \right\rfloor + 1 = \left\lfloor \frac{1251 - 160}{4} \right\rfloor + 1 = 273.$$

Portfolio weights. Let μ be an estimated mean excess-return vector, Σ the `cov1Para` covariance, and $\mathbf{1}$ a vector of ones. We construct three portfolios in each window:

$$\text{Mean-Variance } (\gamma = 1): \quad \mathbf{w}_{\text{MV}} = \frac{1}{\gamma} \Sigma^{-1} \mu, \quad (1)$$

$$\text{Tangency: } \mathbf{w}_{\text{tan}} = \frac{\Sigma^{-1} \mu}{\mathbf{1}^\top \Sigma^{-1} \mu}, \quad (2)$$

$$\text{Equal-weight (EWP): } \mathbf{w}_{\text{EWP}} = \frac{1}{p} \mathbf{1}. \quad (3)$$

The Mean-Variance weights are *not* renormalised to sum to one; the Tangency weights are. The two differ only by the leverage scalar $\mathbf{1}^\top \Sigma^{-1} \mu$, so they share the same risky-asset proportions.

A4: Portfolio Performance

For each test window and portfolio we compute the annualised Sharpe ratio from weekly excess returns,

$$\text{SR}^{\text{ann}} = \sqrt{52} \frac{\bar{r}_p}{\hat{\sigma}(r_p)},$$

`matchingPerformanceAnalytics::SharpeRatio.annualized(scale = 52, geometric = FALSE)`. Tables 1 and 2 report the mean and standard deviation of these ratios across (a) the Full out-of-sample period, (b) the Financial Crisis (06/10/2008–03/08/2011) and (c) the Pandemic (26/02/2020–29/12/2023).

Table 1: Mean-Variance portfolio ($\gamma = 1$): mean and standard deviation of annualised Sharpe ratios across the 273 windows.

Estimator	Full		Crisis		Pandemic	
	Mean	SD	Mean	SD	Mean	SD
Sample	0.5881	5.9739	−1.3206	6.7415	1.7879	9.1497
St-MSh	0.5881	5.9739	−1.3206	6.7415	1.7879	9.1497
D-MSh	0.4540	4.7486	−0.5779	6.0677	0.8014	5.5318
O-LSh	0.4584	4.6680	−0.8870	6.4159	0.3271	5.0471
T-LSh	0.3590	5.1564	−1.9543	6.9767	0.8730	5.8867
Wang	1.1387	6.1079	2.6162	5.1173	0.5545	3.6671
BOP	−0.2139	6.8593	1.1495	5.7339	−1.7032	8.1416
Jorion	0.5578	5.3493	−1.6636	6.0837	1.5921	8.1785
EWP	1.5377	5.3955	2.6972	6.0839	0.9413	4.0546

A5: Interpretation of Results and Final Report

Data and methodology

We study 100 large U.S. companies, drawn at random from the 441 firms that stayed in the S&P 500 for the whole 2000–2023 period, fixed by seed 7. For every company we use its weekly stock return, and from each return we subtract the matching risk-free interest rate, so that we measure the reward investors earn for taking on risk rather than holding cash. This gives 1,251 weeks of “excess return” data per company.

To judge how well each method would have performed in real time, we use a rolling-window backtest: we train on three years (156 weeks), test on the next month (4 weeks), then roll forward by one month and repeat. This produces 273 independent out-of-sample windows from 2003 to 2023, so no result depends on hindsight. In each window we estimate the expected return of

Table 2: Tangency portfolio: mean and standard deviation of annualised Sharpe ratios across the 273 windows.

Estimator	Full		Crisis		Pandemic	
	Mean	SD	Mean	SD	Mean	SD
Sample	0.8136	5.9473	0.4446	6.8583	1.7879	9.1497
St-MSh	0.8136	5.9473	0.4446	6.8583	1.7879	9.1497
D-MSh	0.5272	4.7410	0.0421	6.0958	0.8014	5.5318
O-LSh	0.8336	4.6156	1.5225	6.2921	0.3271	5.0471
T-LSh	0.5701	5.1372	0.4848	7.2359	0.8730	5.8867
Wang	0.8856	6.1499	0.2903	5.7563	0.6193	3.6565
BOP	0.5823	6.8378	0.6125	5.8181	1.3130	8.2151
Jorion	0.9107	5.3004	0.9449	6.2400	1.5921	8.1785
EWP	1.5377	5.3955	2.6972	6.0839	0.9413	4.0546

each stock (the mean vector) and how the stocks move together (the covariance matrix), then form three portfolios: a Mean-Variance portfolio, a Tangency portfolio and an equally weighted benchmark (EWP). We score each by its annualised Sharpe ratio.

Mean-Variance versus Tangency portfolios

In theory the two optimised portfolios should earn the same Sharpe ratio, because the Mean-Variance weights are a scaled version of the Tangency weights and the Sharpe ratio is scale-invariant. Our full-period averages are close but not identical (for example, the sample-mean Tangency portfolio averages 0.81 versus 0.59 for Mean-Variance), and the gap is widest during the Financial Crisis (sample mean: +0.44 Tangency versus −1.32 Mean-Variance). The cause is the Tangency normalisation. Its weights are divided by $\mathbf{1}^\top \Sigma^{-1} \mu$, and whenever that quantity is negative the renormalisation flips the sign of the whole portfolio, turning a losing month into a winning one. The Mean-Variance portfolio keeps the raw sign, so the two diverge precisely in the stressed windows where estimated expected returns are most fragile. Crucially, *neither* optimised portfolio beats the naive equal-weight benchmark over the full period (EWP averages 1.54). With only 156 weeks to estimate 100 means, the optimiser amplifies noise, which is the classic estimation-error result.

Differences among the mean estimators

Over the full sample the best optimised performers are Wang (1.14 Mean-Variance) and, for the Tangency portfolio, Jorion (0.91) and O-LSh (0.83). St-MSh reproduces the sample-mean result exactly in every cell: it multiplies the sample mean by a single positive scalar, which cancels out of the Sharpe ratio, so it offers no improvement here. BOP is the weakest, turning negative over the full period (−0.21 Mean-Variance) and in the Pandemic (−1.70), consistent with its known sensitivity to its fixed shrinkage target.

Robustness differs sharply across regimes. During the Financial Crisis the estimators that shrink toward a central target hold up best. Wang (+2.62 Mean-Variance) and O-LSh (+1.52 Tangency) are the standouts, while shrinking toward zero (St-MSh, identical to the sample mean) fares poorly, and T-LSh posts the single worst crisis loss (−1.95 Mean-Variance). During the Pandemic the picture reverses: the sample mean and Jorion deliver the highest means (around 1.6 to 1.8) but with the largest standard deviations (8 to 9), so high reward is bought with high instability, whereas O-LSh and D-MSh give lower but far steadier results (SD around 5.0 to 5.5). The grand-mean shrinkage estimators (O-LSh, T-LSh, D-MSh) consistently achieve the lowest standard deviations, confirming that their value is risk reduction and stability rather than headline return.

Sector composition and interpretation

The 100 stocks span all 11 GICS sectors (Table 3), with the largest tilts toward Health Care (16%), Information Technology (14%) and Industrials (13%), and only marginal exposure to Materials (1%).

Table 3: GICS sector composition of our 100 selected stocks.

GICS Sector	No. of stocks	Weight (%)
Health Care	16	16
Information Technology	14	14
Industrials	13	13
Consumer Discretionary	12	12
Financials	11	11
Consumer Staples	10	10
Energy	7	7
Real Estate	7	7
Utilities	5	5
Communication Services	4	4
Materials	1	1

This composition helps explain the regime pattern. Our portfolio leans toward defensive, quality sectors. Health Care, Consumer Staples (10%) and Utilities (5%) make up roughly a third of the names, which cushions drawdowns and is consistent with the strong equal-weight benchmark during the crisis. The sizeable cyclical and growth exposure (Information Technology, Industrials, Consumer Discretionary, Financials) drives the high but volatile Pandemic returns of the sample-mean and Jorion portfolios, since these sectors swung most violently in 2020. The grand-mean shrinkage estimators dampen the influence of any single sector's extreme estimated return, which is why they trade away some upside for materially lower volatility in both stressed sub-periods.

Conclusion

For our 100 stocks, shrinkage of the mean vector improves stability but does not let an optimised portfolio overtake the equal-weight benchmark over the full sample, underlining how severe estimation error is when we estimate 100 means from three years of weekly data. Among the optimised methods, Wang and the grand-mean linear estimators (O-LSH, T-LSH) offer the best risk-adjusted trade-off, performing relatively well in the crisis and most steadily in the pandemic, while BOP and pure zero-shrinkage (St-MSH) are the least reliable.

Coursework 2: Shrinkage Generalised Linear Models for Insurance Claim Severity Estimation

Data Description and Preprocessing

We use the French motor third-party liability data: the frequency table `freMTPL2freq.csv` (677,991 policies with 12 risk features) and the severity table `freMTPL2sev.csv` (26,444 individual claims). We aggregate severity to policy level by summing `ClaimAmount` over `IDpol` and inner-join it to the frequency table on `IDpol`. Following the brief, we keep policies with exactly one claim (`ClaimNb == 1`) and a strictly positive amount, which leaves 23,571 eligible policies, and we draw exactly 1,000 of them with `set.seed(7)`. The target is per-policy severity $y_i = \text{ClaimAmount}_i / \text{ClaimNb}_i$, which ranges from €4.1 to €114,760 (mean €2,051, median €1,172) and is strongly right-skewed, which motivates the Gamma family.

We apply bounded log transforms to skewed continuous covariates (**VehPower** capped to $[1, 9]$, **Density** floored at 1, **BonusMalus** capped at 150). **DrivAge** and **VehAge** enter through natural cubic splines ($df = 3$) whose knots are learned on the training fold and projected onto the test fold to prevent leakage. We handle the high-cardinality categoricals (**Region**, **Area**, **VehBrand**) with credibility-weighted leave-one-out target encoding (smoothing constant $k = 20$), and we standardise all engineered predictors using training-fold means and standard deviations only. For a Gamma GLM with a log link the linear predictor is $\hat{\eta}_i = x_i^\top \hat{\beta}$ and the fitted severity is $\hat{y}_i = \hat{\mu}_i = \exp(\hat{\eta}_i)$. All results below are reproducible from the two provided datasets together with the R script we submit alongside this report.

B1: Out-of-Sample Performance Evaluation

We run a 5-fold cross-validation repeated 100 times. Within each repetition we partition the 1,000 policies into five folds; each fold serves once as the 20% test set while the other 80% trains the models, giving 500 train/test evaluations per model. We re-randomise the fold partition per repetition (seed 123 + repeat) so that all nine models see an identical split within a run, making the comparison paired. For every run we record twelve metrics: RMSE, MAE, their 99% and 95% winsorised variants, Gamma deviance, raw and normalised Gini, the actual-to-expected ratio, computation time and the number of solver iterations. The penalised models (Ridge $\alpha = 0$, Elastic Net $\alpha = 0.5$) select λ by an inner 5-fold CV; the shrinkage GLMs use `savvy_glm.fit2` with the relevant model class.

Table 4: Average absolute out-of-sample metrics over 500 runs. Smaller is better for RMSE, MAE, the winsorised errors, Dev, CT and NoIt; larger for Gini; closer to 1 for AE.

Model	RMSE	MAE	RMSE ₉₉	RMSE ₉₅	MAE ₉₉	MAE ₉₅	Dev	Gini _{raw}	Gini _{norm}	AE	CT(s)	NoIt
GLM2	6012.5	1899.3	2838.4	1475.9	1545.5	1201.3	329.8	-0.0151	-0.0270	1.0415	0.0117	17.3
Ridge	5971.1	1849.4	2772.2	1353.2	1495.6	1152.4	309.6	0.0237	0.0361	1.0516	1.6493	581.3
ElasticNet	5975.7	1859.7	2777.3	1365.6	1505.9	1162.7	312.4	0.0072	0.0052	1.0481	1.5093	798.7
St	6007.0	1866.9	2828.2	1447.9	1513.0	1169.0	331.1	-0.0150	-0.0268	1.0662	0.0695	19.7
DSh	5986.5	1847.8	2799.7	1400.8	1494.0	1150.3	321.5	-0.0056	-0.0111	1.0688	0.0474	5.6
SR	6072.2	1999.0	2927.8	1637.3	1645.2	1289.6	372.4	-0.0256	-0.0356	0.9924	0.0515	7.0
GSR	6024.1	1907.7	2854.0	1492.2	1553.6	1207.1	330.4	-0.0065	-0.0135	1.0404	0.0517	7.8
LW	6134.2	2231.5	3072.2	1924.7	1877.7	1518.4	334.8	0.0502	0.0775	0.8857	0.0384	9.1
QIS	6026.0	1989.5	2915.5	1584.9	1635.7	1275.4	331.0	0.0080	0.0115	1.0142	0.0378	8.5

Table 5: Average relative out-of-sample performance versus GLM2 (mean of per-run ratios over 500 runs). The error, Dev, CT and NoIt columns are ratios (< 1 beats GLM2); Gini and AE are shown in absolute terms.

Model	RMSE	MAE	RMSE ₉₉	RMSE ₉₅	MAE ₉₉	MAE ₉₅	Dev	Gini _{raw}	Gini _{norm}	AE	CT $\times$	NoIt $\times$
GLM2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	-0.0151	-0.0270	1.0415	1.00	1.00
Ridge	0.9872	0.9725	0.9754	0.9181	0.9680	0.9605	0.9395	0.0237	0.0361	1.0516	169.35	41.37
ElasticNet	0.9889	0.9789	0.9774	0.9264	0.9748	0.9687	0.9490	0.0072	0.0052	1.0481	151.38	54.02
St	0.9978	0.9817	0.9960	0.9812	0.9789	0.9733	1.0023	-0.0150	-0.0268	1.0662	6.58	1.20
DSh	0.9920	0.9709	0.9854	0.9498	0.9667	0.9583	0.9746	-0.0056	-0.0111	1.0688	4.88	0.40
SR	1.0146	1.0532	1.0337	1.1093	1.0652	1.0732	1.1178	-0.0256	-0.0356	0.9924	5.32	0.50
GSR	1.0039	1.0049	1.0059	1.0107	1.0052	1.0046	1.0033	-0.0065	-0.0135	1.0404	5.27	0.54
LW	1.0440	1.1844	1.0884	1.3033	1.2162	1.2633	1.0335	0.0502	0.0775	0.8857	3.88	0.66
QIS	1.0020	1.0423	1.0275	1.0746	1.0583	1.0627	1.0029	0.0080	0.0115	1.0142	3.85	0.60

Relative to the GLM2 baseline, every penalised and shrinkage estimator lowers tail-sensitive error. Ridge cuts RMSE₉₅ by 8.2% and deviance by 6.1%, with Elastic Net close behind. The pure shrinkage GLMs (St, DSh, GSR) sit between GLM2 and the penalised models on accuracy but at far lower cost: Ridge and Elastic Net cost roughly $169\times$ and $151\times$ the GLM2 fitting

time because of the inner CV over λ , whereas DSh, St and GSR cost only 5 to 7 $\times$, and DSh converges in 0.40 $\times$ the GLM2 iterations. Discrimination is weak for all models, as expected, since individual claim size in a single accident year carries little rankable signal; LW achieves the highest Gini but the worst calibration (AE = 0.886, about 11% aggregate over-shrinkage), while SR is best calibrated (AE = 0.992).

B2: Model Comparison and Selection

No single estimator dominates every criterion, so selection is a trade-off. On the accuracy metrics that matter most for severity pricing (RMSE, the winsorised errors and Gamma deviance), Ridge is the clear leader: it posts the lowest RMSE (5,971), the lowest RMSE₉₅ (0.918 $\times$ GLM2) and the lowest deviance (0.940 $\times$ GLM2), while keeping a positive if modest Gini and an AE of 1.05. Elastic Net is marginally behind on every accuracy column. The shrinkage GLMs trail the penalised pair slightly on accuracy but win decisively on efficiency. In particular, DSh attains RMSE 0.992 $\times$, MAE 0.971 $\times$ and deviance 0.975 $\times$ GLM2, recovering most of Ridge’s accuracy gain, at only 4.9 $\times$ GLM2 runtime versus Ridge’s 169 $\times$, and with the fewest iterations of any method. LW maximises Gini but is poorly calibrated, and SR and QIS are well calibrated but less accurate. We prioritise predictive accuracy and goodness of fit for a pricing model, so we select Ridge as the best overall performer, and we note DSh as the efficiency-optimal choice where models must be refitted frequently at scale, since it delivers near-Ridge accuracy at roughly one thirty-fifth of the computational cost.

B3: In-Sample Performance Analysis

We refit the selected Ridge model ($\alpha = 0$, $\lambda_{\min} = 2.50$) on the full 1,000 policies using the identical pre-processing pipeline, then compare the fitted severities $\hat{y}_i$ to the observed y_i .

Table 6: In-sample fit of the selected Ridge model on all 1,000 policies.

RMSE	6,568	Aggregate AE ratio	1.0378
MAE	1,827	Mean: actual / fitted	2,051 / 1,977
Gamma deviance	1,400	Median: actual / fitted	1,172 / 1,981

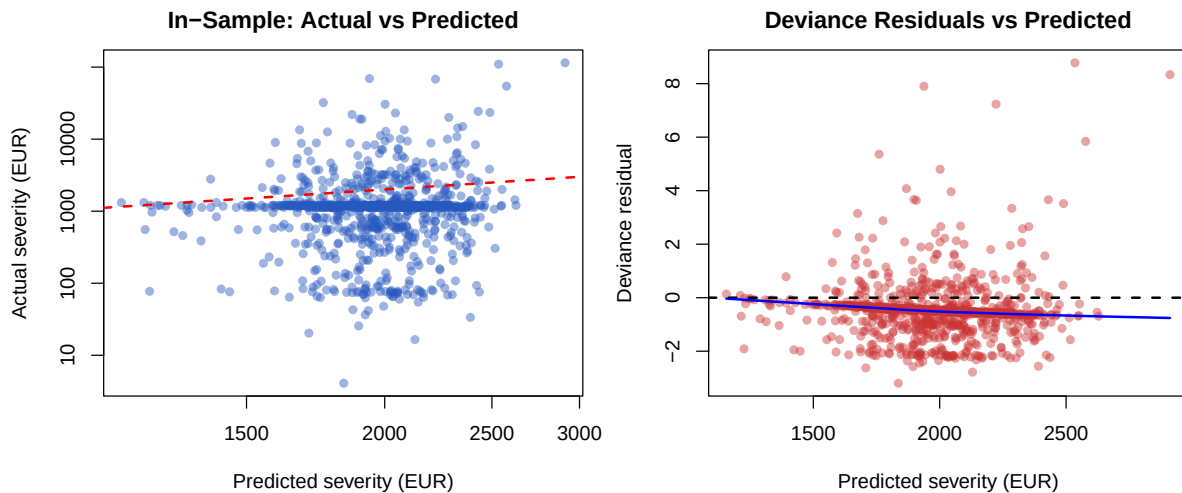


Figure 1: Left: in-sample actual versus fitted severity (log-log); the dashed line is perfect prediction. Right: Gamma deviance residuals versus fitted values with a lowess trend.

The aggregate AE ratio of 1.038 shows the portfolio is priced to within about 4% on total severity, which is the property an insurer cares about most for reserving, and the fitted mean (€1,977) tracks the actual mean (€2,051) closely. The model is deliberately a shrinkage estimator, so the fitted values occupy a much narrower band than the data (fitted median €1,981 versus actual median €1,172). It therefore over-predicts typical small claims and cannot reach the extreme tail, so claims above €30,000 in Figure 1 sit far above the 45° line. The residual panel shows the classic Gamma funnel: residuals are centred near zero across most of the fitted range, which indicates good mean calibration, but they fan out at high predictions. This confirms that a handful of catastrophic claims, rather than systematic bias, drive the bulk of the squared error. This is the expected behaviour of a regularised severity model: it trades individual tail accuracy for stable, well-calibrated aggregate pricing and strong out-of-sample generalisation, which is why it outperformed the unpenalised GLM2 in B1.

Conclusion

Across 500 out-of-sample runs, regularisation consistently improves on the unpenalised GLM2 for Gamma severity: every penalised and shrinkage estimator lowers tail-sensitive error, and Ridge delivers the strongest accuracy and goodness of fit while DSh recovers almost the same gain at a fraction of the computational cost. We therefore select Ridge as the best overall model and note DSh as the efficient alternative for large-scale refitting. The in-sample analysis confirms that our chosen model is well calibrated in aggregate, pricing total severity to within about 4%, and that its residual error is driven by a small number of extreme claims rather than systematic bias, which is the expected and desirable behaviour of a shrinkage estimator built for stable pricing.